

80W Solar Panel for Battery Plus Datasheet

Overview

Description

This 80 W solar panel is specially selected to meet the needs of the Battery Plus system making it the perfect accessory for your green solutions or ones in remote areas. It is based on monocrystalline solar cells and has an expected life of more than 25 years. With the included mounting kit, you can build and power an off-grid deployment scenario. Using the **WisGate Edge Pro** gateway in a combination with the **Battery Plus** and the **Solar Panel**, you are getting reliable network coverage without power loss.

Features

- Peak power: 80 W
- Open circuit voltage (Voc): 21.8 V
- Max. power voltage (Vmp): 18.24 V
- Max. power current (Imp): 4.39 A
- Dimensions: 730 x 670 x 30 mm
- Operating temperature: -40° C to +85° C

Specifications

Overview



Figure 1: Overview

Parameter	Value
Dimensions (l/w/t)	730 x 670 x 30 mm
Solar cell type	Monocrystalline silicon
No. of cells	36
Front cover (material/thickness)	Low-iron tempered glass / 3.2 mm
Encapsulant (material)	Ethylene vinyl acetate (EVA)
Frame (material/color)	Anodized aluminum alloy / silver

Parameter	Value
Junction box protection degree	IP67

Hardware

Electrical Characteristics

Parameter	Value
Model	YXP-80 W
Peak power (Pm)	80 W ($\pm 3\%$)
Open circuit voltage (Voc)	21.8 V
Max. power voltage (Vmp)	18.24 V
Max. power current (Imp)	4.39 A
Short circuit current (Isc)	4.65 A
Standard test conditions	Cell temperature 25° C ($\pm 2^\circ$ C); Irradiance 1000° CW/m2; Standard solar spectral distribution AM1.5
Coefficient current TK	0.0576% / °C
Coefficient voltage TK	-0.340% / °C
Coefficient power TK	-0.398% / °C

Mechanical Characteristics

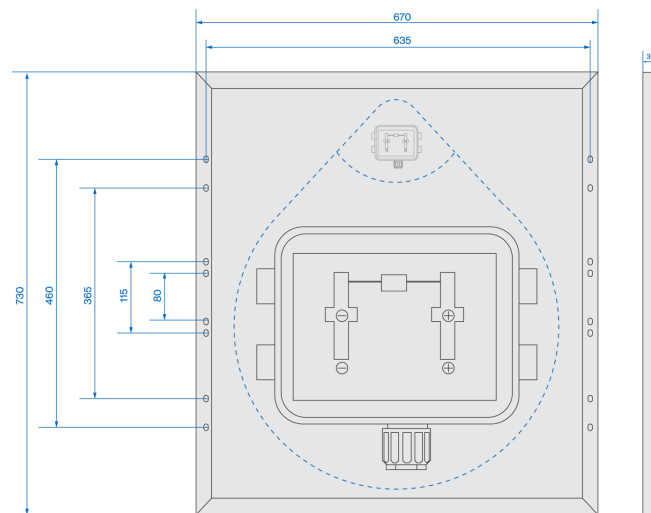


Figure 2: Solar panel dimensions

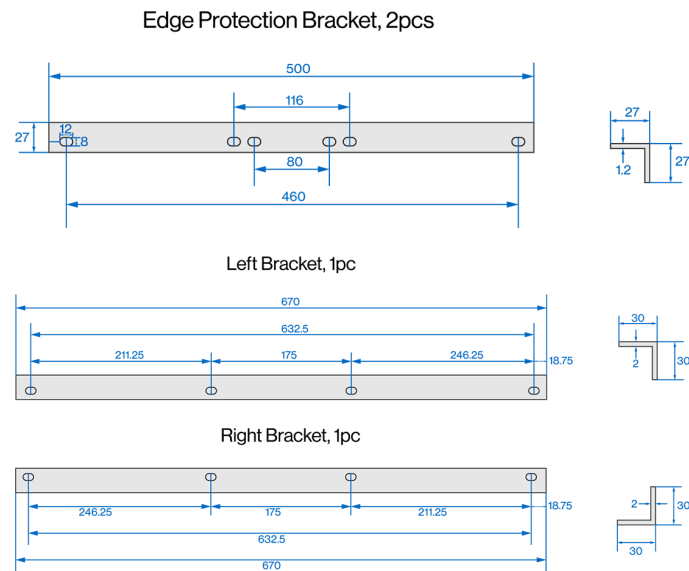


Figure 3: Mounting elements

Environment Characteristics

Parameter	Value
Max. static load, front (snow/wind)	5400 Pa
Max. static load, back (wind)	2400 Pa
Max. hailstone impact (diameter/velocity)	25 mm / 23 m/s
Operating temperature	-40° C to +85° C

i NOTE

This guide covers winter operation for RAK solar and battery systems. For deployment-specific instructions, refer to [Winter Deployment Guide](#).